Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Cross #7 (Formerly Larry B. Peterson Dairy Farms #2)

Physical address of dairy:

Number and Street	City	County	Zip Code
20775 W American AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1979

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X170-X030-XXXX

B. OPERATORS

Ahlem, Mark

Operator name: Ahlem, Mark

Telephone no.: (209) 668-0867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Peterson, Mike or Randy

Legal owner name: Peterson, Mike or Randy

Telephone no.: (209) 667-5888 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
20949 Bloss AVE	Hilmar	CA	95324

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	208	0	0	0	0
Maximum number	0	208	0	0	0	0
Average number	0	208	0	0	0	0
Avg live weight (lbs)	0	1,000	0	0		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 2,663.19 *tons per reporting period*

Total nitrogen from manure: 37,960.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 26,572.00 *lbs per reporting period*

Total phosphorus from manure: 5,010.72 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 47,829.60 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,664,000 *gallons*

Total nitrogen generated: 5,492.88 *lbs*

Total phosphorus generated: 1,358.98 *lbs*

Total potassium generated: 5,666.48 *lbs*

Total salt generated: 46,112.47 *lbs*

	5,193,000	*gallons applied*
+	0	*gallons exported*
-	2,529,000	*gallons imported*
=	2,664,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
03/02/2023	Process wastewater *Imported WWS*	255,000.00 *gal*	297.00	65.50	418.00		3,640
04/15/2023	Process wastewater *Imported WWS*	240,000.00 *gal*	549.00	43.80	1,010.00		5,860
07/22/2023	Process wastewater *Imported WWS*	510,000.00 *gal*	230.00	66.70	260.00		1,440
08/25/2023	Process wastewater *Imported WWS*	744,000.00 *gal*	230.00	66.70	260.00		1,440
09/03/2023	Process wastewater *Imported WWS*	390,000.00 *gal*	230.00	66.70	260.00		1,440
09/18/2023	Process wastewater *Imported WWS*	390,000.00 *gal*	230.00	66.70	260.00		1,440

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	5,635.50	1,359.25	7,325.49	43,924.41
Total imports for all materials	5,635.50	1,359.25	7,325.49	43,924.41

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	27	27	2	both	X045-X170-X030-XXXX
Totals for areas that were used for application	27	27	2		
Totals for areas that were not used for application					
Land application area totals	27	27	2		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	405.00 *ton*	As-is		69.2	7,100.00	1,400.00	12,600.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	213.00	42.00	378.00	896.28

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	634.56 *ton*	As-is		59.8	5,600.00	1,000.00	9,500.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	263.22	47.00	446.54	1,303.81

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/20/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	103.79	7.41	25.95	0.00	400.00 *ton*
Application event totals		103.79	7.41	25.95	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.68	0.93	7.16	430.82	288,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	286.81	4,316,166.00 *gal*
Application event totals		23.68	0.93	7.16	717.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.67	3.46	23.51	133.52	120,000.00 *gal*
Application event totals		25.67	3.46	23.51	133.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS-Imported	Process wastewater	23.41	5.16	32.94	286.88	255,000.00 *gal*
Application event totals		23.41	5.16	32.94	286.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS-Imported	Process wastewater	40.72	3.25	74.92	434.68	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	228.06	3,432,024.00 *gal*
Application event totals		40.72	3.25	74.92	662.74	

Field 1 - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	215.05	3,236,166.00 *gal*
Application event totals		0.00	0.00	0.00	215.05	

Field 1 - 06/14/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	288.86	4,346,946.00 *gal*
Application event totals		0.00	0.00	0.00	288.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.57	19.55	76.25	486.61	960,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	266.55	4,011,255.00 *gal*
Application event totals		65.57	19.55	76.25	753.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS-Imported	Process wastewater	36.25	10.51	40.98	226.98	510,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	236.93	3,565,485.00 *gal*
Application event totals		36.25	10.51	40.98	463.91	

Field 1 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	216.30	3,255,066.00 *gal*
Application event totals		0.00	0.00	0.00	216.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.43	10.27	40.03	255.47	504,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	220.66	3,320,703.00 *gal*
Application event totals		34.43	10.27	40.03	476.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	195.52	2,942,379.00 *gal*
Application event totals		0.00	0.00	0.00	195.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported WWS	Process wastewater	52.89	15.34	59.79	331.13	744,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	200.21	3,012,974.00 *gal*
Application event totals		52.89	15.34	59.79	531.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS-Imported	Process wastewater	27.72	8.04	31.34	173.58	390,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	181.03	2,724,316.00 *gal*
Application event totals		27.72	8.04	31.34	354.61	

Field 1 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.10	16.13	62.91	401.45	792,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	167.52	2,520,996.00 *gal*
Application event totals		54.10	16.13	62.91	568.97	

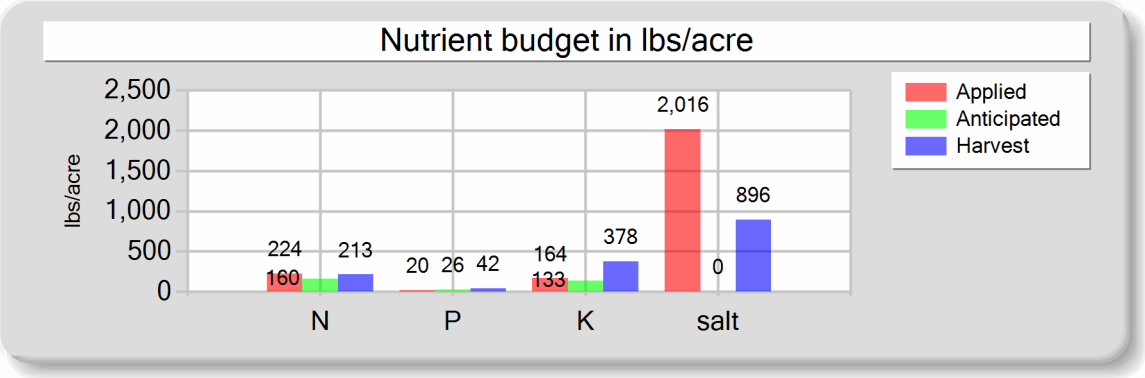
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS-Imported	Process wastewater	27.72	8.04	31.34	173.58	390,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	149.26	2,246,211.00 *gal*
Application event totals		27.72	8.04	31.34	322.84	

B. NUTRIENT BUDGET

Field 1 - 10/20/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.79	7.41	25.95	0.00
Process wastewater	113.47	12.79	138.53	1,285.91
Fresh water	0.00	0.00	0.00	729.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.26	20.21	164.48	2,015.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	213.00	42.00	378.00	896.28
Nutrient balance	11.26	-21.79	-213.52	1,119.55
Applied to removed ratio	1.05	0.48	0.44	2.25

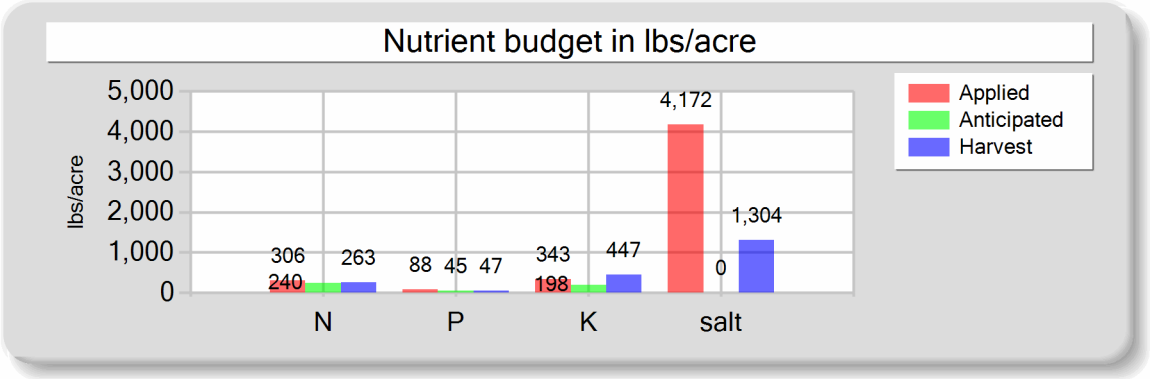
Fresh water applied	
10,984,356.00	*gallons*
404.52	*acre-inches*
14.98	*inches/acre*

Process wastewater applied	
903,000.00	*gallons*
33.25	*acre-inches*
1.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/14/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	298.69	87.88	342.65	2,048.79
Fresh water	0.00	0.00	0.00	2,122.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.69	87.88	342.65	4,171.65
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	263.22	47.00	446.54	1,303.81
Nutrient balance	42.46	40.88	-103.89	2,867.84
Applied to removed ratio	1.16	1.87	0.77	3.20

Fresh water applied	
31,946,331.00	*gallons*
1,176.47	*acre-inches*
43.57	*inches/acre*

Process wastewater applied	
4,290,000.00	*gallons*
157.99	*acre-inches*
5.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Separated Solids

Sample and source description: 2022 Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,400.00	600.00	2,100.00							
DL	100.00	100.00	100.00							

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,300.00	3,600.00							55.70
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,500.00	2,000.00	3,100.00	6,100.00	1,900.00	1,000.00	2,100.00	408.00		32.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	266.00	176.00	0.00	0.00	10.40	80.40								4,090.00	4,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ1

Sample and source description: Imported WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	297.00	176.00	0.00	0.00	65.50	418.00								436.00	3,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	692.00	333.00	0.00	0.00	93.20	634.00								5,620.00	3,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ2

Sample and source description: Imported WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	381.00	0.00	0.00	43.80	1,010.00								7,770.00	5,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	566.00	378.00	0.00	0.00	45.50	1,010.00								7,710.00	5,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Imported WWSQ3

Sample and source description: Imported WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	230.00	123.00	0.00	0.00	66.70	260.00								2,610.00	1,440
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	221.00	123.00	0.00	0.00	65.90	257.00	79.60	57.00	62.70	1,120.00	0.00	77.60	101.00	2,630.00	1,640
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	428.00	286.00	0.00	0.00	34.70	489.00								5,190.00	3,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Lateral 6

Sample description: Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,100.00	1,400.00	12,600.00		9.70
DL	100.00	100.00	100.00		0.01

Field 1 - 06/14/2023: Corn, silage

Field 1 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,000.00	9,500.00		6.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

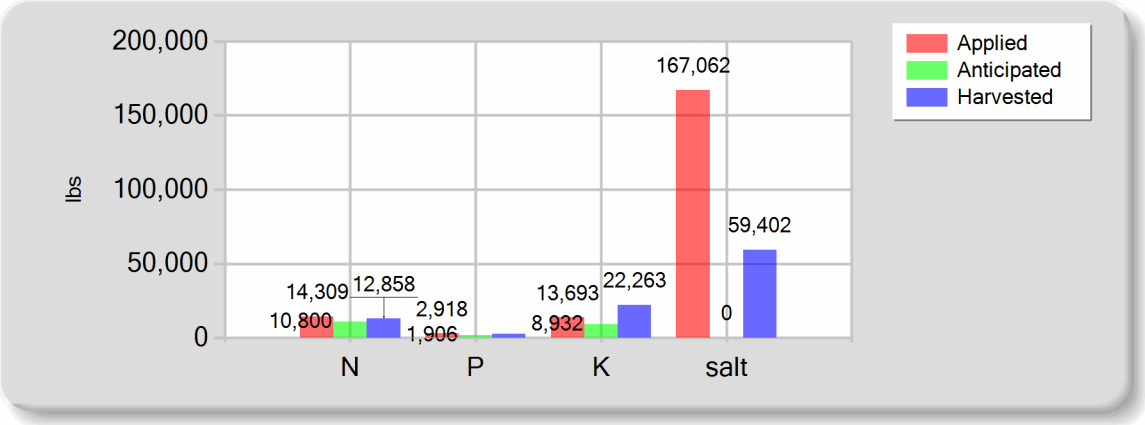
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

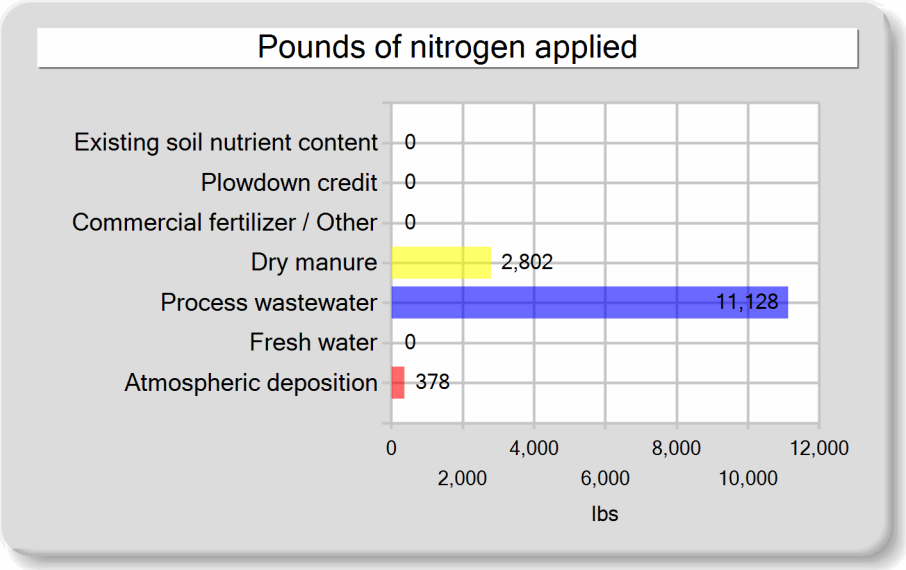
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

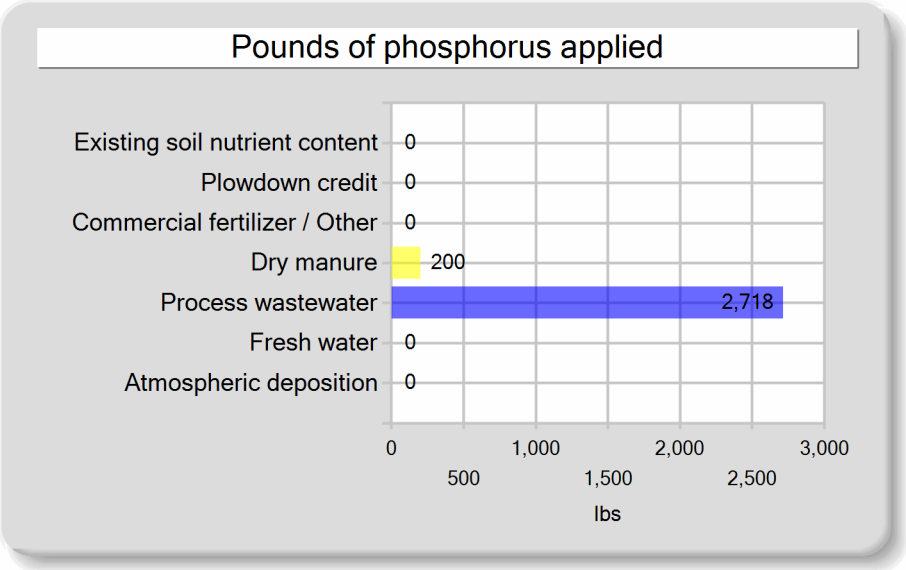
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	2,802.24	200.16	700.56	0.00
Process wastewater	11,128.38	2,718.23	12,991.97	90,036.88
Fresh water	0.00	0.00	0.00	77,025.17
Atmospheric deposition	378.00	0.00	0.00	0.00
Total nutrients applied	14,308.62	2,918.39	13,692.53	167,062.04
Anticipated crop nutrient removal	10,800.00	1,906.20	8,931.60	0.00
Actual crop nutrient removal	12,858.07	2,403.12	22,262.64	59,402.41
Nutrient balance	1,450.55	515.27	-8,570.11	107,659.63
Applied to removed ratio	1.11	1.21	0.62	2.81

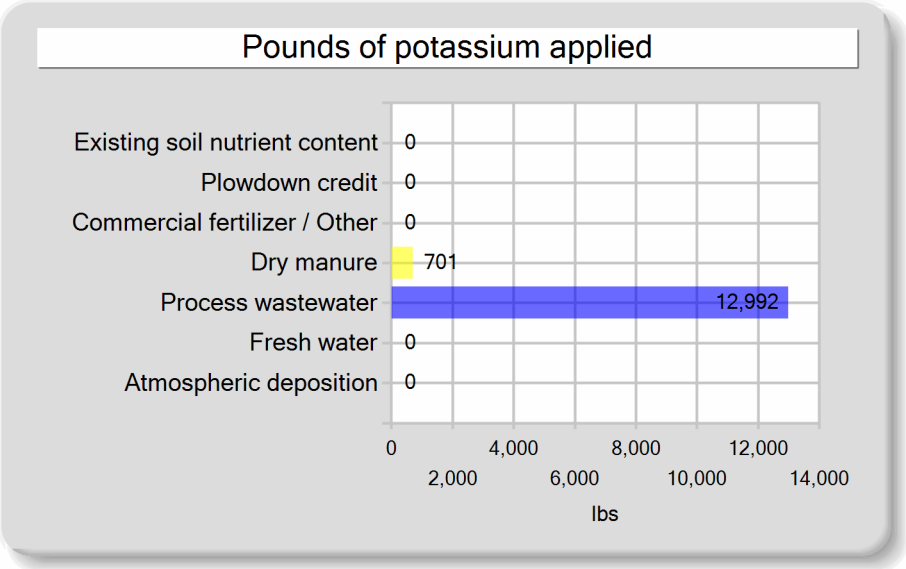
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

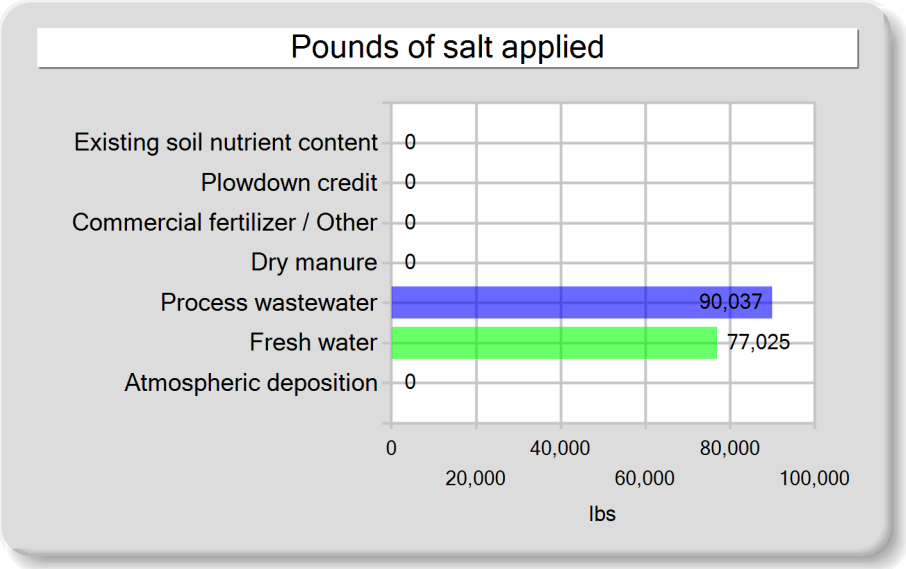


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Dry cows only housed onsite.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mike Peterson Mike Peterson (May 27, 2024 15:53 PDT)	Mark Ahlem Mark Ahlem (May 28, 2024 11:27 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mike or Randy Peterson	Mark Ahlem
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 27, 2024 5/27/24	May 28, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #7
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3291013A	Electrical Conductivity (EC)	EPA 120.1 and	0.820	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	525	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DC #7
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To:
FR
Sampled By: SB
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/10	2:00	Domestic well	DW#1	15.6°C		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FPAG				
2nd		DENELE ANALYTICAL, INC.	10-18-23 9:45			T3291013

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Double Cross #7

Final Audit Report 2024-05-28

Created:	2024-05-21
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAnG9-BCwELZAc7EcGRRiWCnWVx3SwguMA

"2023 Annual Report Double Cross #7" History

Document created by Kristen Jones (kjones@fragservices.com)
2024-05-21 - 3:51:09 PM GMT

Document emailed to lbpetersondairy@gmail.com for signature
2024-05-21 - 3:52:10 PM GMT

Email viewed by lbpetersondairy@gmail.com
2024-05-21 - 3:52:16 PM GMT

Signer lbpetersondairy@gmail.com entered name at signing as Mike Peterson
2024-05-27 - 10:53:19 PM GMT

Document e-signed by Mike Peterson (lbpetersondairy@gmail.com)
Signature Date: 2024-05-27 - 10:53:21 PM GMT - Time Source: server

Document emailed to markahlem@gmail.com for signature
2024-05-27 - 10:53:23 PM GMT

Email viewed by markahlem@gmail.com
2024-05-28 - 6:37:40 AM GMT

Signer markahlem@gmail.com entered name at signing as Mark Ahlem
2024-05-28 - 6:27:54 PM GMT

Document e-signed by Mark Ahlem (markahlem@gmail.com)
Signature Date: 2024-05-28 - 6:27:56 PM GMT - Time Source: server

Agreement completed.
2024-05-28 - 6:27:56 PM GMT